

Employee Attrition Prediction Using Machine Learning

Executive Summary for HR Director

Project Overview

Employee attrition is a major challenge for organizations because replacing experienced employees requires significant time, cost, and effort. The objective of this project was to analyse employee data and develop a machine learning model capable of identifying employees who are at a higher risk of leaving the organization. The analysis considered multiple factors, including job role, department, overtime, business travel, work-life balance, monthly income, and years at the company.

Key Findings

The analysis revealed that employee attrition is concentrated within specific departments and job roles rather than being evenly distributed across the organization. The **Sales department** recorded the highest attrition rate (**20.63%**), while **Sales Representatives** experienced the highest turnover among all job roles (**39.76%**). Employees reporting poor work-life balance and those in their first two years of employment were considerably more likely to leave the company.

The machine learning analysis also demonstrated that **salary alone does not explain employee attrition**. Instead, factors such as overtime, business travel, job role, career progression, and work-life balance have a stronger influence on an employee's decision to leave.

Model Performance

Three machine learning models were developed and compared: **Logistic Regression**, **Random Forest**, and **Gradient Boosting**. Among these models, **Logistic Regression** achieved the best overall performance, recording the highest Recall (**0.617**), F1-Score (**0.439**), and ROC-AUC Score (**0.799**). Because identifying employees who are likely to resign is more important than minimizing false alarms, Logistic Regression was selected as the most appropriate model for this business problem.

Business Recommendations

Based on the analysis, HR should prioritize retention initiatives for employees working in the **Sales department**, particularly **Sales Representatives**, as well as other high-risk roles identified by the model. The organization should also reduce excessive overtime, promote a healthier work-life balance, and strengthen onboarding and mentoring programs during employees' first two years of service. Regular career development discussions and employee engagement initiatives can further improve long-term retention.

Project Conclusion

This project demonstrates that machine learning can help HR identify employees at risk of leaving and support proactive retention strategies. While the model provides valuable insights, it should be used alongside human judgment, as factors such as organizational culture and personal circumstances are not included in the dataset. Combining predictive analytics with effective HR initiatives can improve employee retention and workforce stability.